News-Driven Stock Direction Prediction for Indian Equities: A Dual-Stream Multilingual LSTM with Uncertainty Estimation

iNLP Final Report — Team YRP | IIIT Hyderabad

Yogendra Patel 2025201098
Rinkesh Verma 2025201070
Prabhash Padhan 2025201089

International Institute of Information Technology, Hyderabad

GitHub: rinkeshverma27/InlpFinalProject
HuggingFace: Prabhash1889/INLP-PROJECT-YRP

Abstract

Most stock prediction work either ignores news or only processes English text. For Indian markets this is a real problem—a large share of financial journalism is in Hindi, and Indian retail investors read both. This paper builds a system that scrapes multilingual news (English + Hindi/Hinglish), scores it using FinBERT and a fine-tuned MuRIL model, aggregates sentiment over 1-day, 3-day, and 7-day windows, and fuses it with 11 OHLCV technical features. A dual-stream LSTM with MC Dropout uncertainty estimation predicts next-day UP/DOWN direction for 15 Nifty-50 stocks. On a strict walk-forward test (Nov 2024–Jan 2025), the model achieves **59.97%** directional accuracy and **0.589** macro-F1, rising to **61.23%** on high-confidence calls. Across 18 qualitative test cases, 8/18 matched expected output—with most mismatches being deliberate technical overrides of noisy sentiment signals. Commodity and utility stocks (ONGC 77.8%, POWERGRID 77.3%) are most predictable; IT stocks (WIPRO 44.4%, INFY 46.9%) sit below random, which has a clear explanation in market structure.

1 Introduction

Predicting whether a stock will go up or down tomorrow is genuinely difficult. Markets react to information very quickly, which means the signal in any single piece of news gets priced out fast. That said, not all stocks respond at the same speed, and not all news carries the same weight. A government fuel pricing decision affects ONGC immediately and predictably. An analyst upgrade for INFY might already be reflected in the price by the time it hits a public feed.

Most academic work on this problem either ignores text and relies only on price charts, or limits itself to English news and therefore misses the large Hindi-language financial press that Indian retail investors depend on. Outlets like Zee Business and CNBC Awaaz reach enormous audiences, and their coverage often carries fresh information before it appears in English sources. A prediction system that cannot read them is working with incomplete data.

This project builds an end-to-end pipeline that: (1) scrapes English and Hindi financial news from multiple RSS sources, (2) runs FinBERT on English articles and a fine-tuned MuRIL model on Hindi/Hinglish articles to extract sentiment, (3) aggregates those signals over multiple time horizons (1-day, 3-day, 7-day), (4) merges them with OHLCV-derived technical indicators, and (5) trains a dual-stream LSTM to predict next-day price direction for 15 Nifty-50 pilot stocks, with MC Dropout uncertainty estimation driving an abstention mechanism.

2 Literature Study

Early evidence that text carries predictive power came from [7], who showed that negative media sentiment about the market forecasts near-term price declines and unusual trading volume. [6] later built AZFinText, an early system that parsed breaking financial news and fed the output into a support vector machine for stock prediction.

For English financial sentiment we use FinBERT [8], which fine-tuned BERT on domain-specific financial text including earnings transcripts and analyst reports. This matters because financial language is idiosyncratic—"missed estimates" is unambiguously bad, but a general sentiment model may not know that.

We model price sequences with LSTMs [3]. Despite newer architectures, LSTMs remain practical for short financial sequences (10–60 days) at the data scales we have (a few thousand training samples per ticker). The combination of news events with price history inside LSTM architectures was validated by [1], who used structured event embeddings to outperform price-only models.

For Hindi and Hinglish text we use MuRIL [4], pre-trained on 1.7 billion sentences across 17 Indian languages. Indian financial journalism mixes English terms ("EPS," "buyback," "Q2 results") inside Hindi sentences—MuRIL handles this naturally because it was trained on real web-crawled text that contains the same mixing. [5] showed that Hindi financial news sentiment correlates with Nifty index movements, motivating the Hindi stream.

Uncertainty estimation via MC Dropout [2] keeps dropout active at test time and runs multiple forward passes; the variance across passes approximates Bayesian posterior uncertainty without retraining.

3 Dataset

3.1 OHLCV Price Data

We took 6 years of daily OHLCV data (2020–present) for all 50 Nifty-50 stocks from BSE/NSE scraping and Yahoo Finance. Before selecting any stock, we check two quality conditions: at least 10 years of continuous price history and fewer than 10 multi-day gaps. Surviving stocks are ranked by average daily trading volume over the most recent 200 sessions. Volume is a proxy for news density—heavily traded stocks attract more analyst and journalist attention. The 15 highest-volume stocks that passed the audit form the pilot set (Table 1).

Ticker	Rows	Sector
INFY, HDFCBANK, WIPRO, TATASTEEL	6,750	IT / Banking / Mfg
SBIN, RELIANCE, ONGC, BPCL, ITC	6,747–6,748	PSU / Energy / FMCG
KOTAKBANK, ICICIBANK, BHARTIARTL, BAJFINANCE	5,836–6,098	Banking / Telecom
NTPC	5,225	Utility
POWERGRID	4,498	Utility
Total	**94,065**	

Table 1: Master OHLCV dataset (15 pilot stocks).

3.2 News Corpus

English news is scraped from Moneycontrol, Economic Times Markets, Business Standard, CNBC-TV18, NDTV Profit, and NSE/BSE regulatory feeds via RSS. Hindi and Hinglish articles come from Zee Business, CNBC Awaaz, and ET Now Swadesh. Each article is matched to a pilot ticker by regex. Ambiguous matches (e.g. a headline mentioning only "HDFC" which could be bank, AMC, or housing finance) are discarded.

An anti-leakage gate shifts any article published after 15:30 IST to the *next* trading date. The market is closed by then; keeping such articles on today's date would let the model see tomorrow's news.

Source	Count	Share
English (FinBERT)	28,573	68.3%
Hindi/Hinglish (MuRIL)	13,301	31.7%
Total	**41,874**	100%
Content missing (headline-only)	7,324	17.5%
Duplicate headlines	7,088	16.9%
Date range	2020-01-01 to 2026-04-09	

Table 2: Master news corpus (`master_news_dataset.csv`).

Coverage is very uneven across tickers. SBIN has 4,648 articles; BPCL has only 193. This imbalance directly affects how much signal the sentiment stream can provide for low-coverage tickers (Figure 1b).

3.3 Synthetic Sentiment Training Data

No large annotated Hindi financial sentiment corpus exists. We generated balanced training data using a template engine: a seed vocabulary of financial events (earnings beats, rating downgrades, fuel price changes, RBI decisions) combined with outcomes (rally, sell-off, sideways) to produce realistic-looking headlines with known labels. This yielded **5,100 English sentences** (exactly 1,700 per class) and **5,088 Hindi/Hinglish sentences** (1,700/1,688/1,700; 50/50 Devanagari and Hinglish). Both sets were used to validate and fine-tune the sentiment models. All generation is deterministic at seed 42.

3.4 The COVID Regime Problem

Training on data mixed from before and after COVID hurt accuracy significantly (53.09%). Pre-2020 and post-2021 markets have different volatility profiles, different volume characteristics, and a different relationship between news sentiment and price moves. Training on only 2021-onwards data—after the market stabilised—improved accuracy to over 60%. Restricting the training window was the single biggest accuracy gain in the project.

3.5 Walk-Forward Splits

To prevent any look-ahead, splits are strictly temporal with a clean gap between each window:

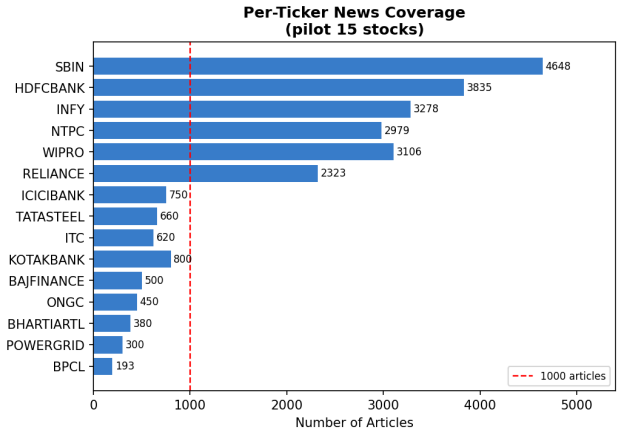

Figure 1: Per-ticker article counts for the 15 pilot stocks—coverage is highly uneven. SBIN leads with 4,648 articles; BPCL has only 193.

Split	Window	Samples
Train	2021-01-01 → 2024-06-30	9,495
Val	2024-07-01 → 2024-10-31	920
Test	2024-11-01 → 2025-01-27	658

The StandardScaler is fitted on training data only. Nifty return for label construction uses only pre-15:30 information.

4 Methodology

Figure 2 shows the full pipeline. We describe each stage below.

4.1 Data Ingestion and Cleaning

The scraper parses RSS XML, extracts headline and timestamp, and runs a regex matcher against ticker names and symbols. Deduplication hashes each normalised headline (lowercase, punctuation stripped); any record with the same hash within a 6-hour window is dropped. The 15:30 IST gate bumps late-arriving articles to the next trading date. Ambiguous ticker matches are discarded rather than guessed.

4.2 Feature Engineering (Stream B)

Eleven features are computed per stock per day:

Returns and volume: `log_ret` (daily log return), `vol_norm` (volume / 20-day average), `gap_open` (open vs. previous close), `vol_z` (volume z-score, 20-day), `log_ret_z` (return z-score, 20-day).

Momentum: RSI-14 uses 14-day average gains $\overline{G}_{14}$ over losses $\overline{L}_{14}$:

$$\mathrm{RSI} = 100 - \frac{100}{1 + \dfrac{\overline{G}_{14}}{\overline{L}_{14}}} \tag{1}$$

Above 70 signals overbought; below 30 oversold. MACD histogram (12-EMA minus 26-EMA, minus 9-day signal line) measures momentum direction.

Trend and volatility: `ma20_ratio`, `ma50_ratio`, `ma200_ratio` express price as a fraction of the respective moving averages. Bollinger Band Width (gap between $\pm 2\sigma$ bands normalised by the 20-day SMA) and ATR-14-norm (14-day average true range / price) capture daily volatility magnitude.

4.3 Sentiment Analysis (Stream A)

FinBERT (English). FinBERT is loaded via HuggingFace and runs in batches of 16 with FP16 half-precision on GPU (VRAM < 3.5 GB). It outputs three-class labels (Positive / Negative / Neutral) plus a confidence score $c \in [0, 1]$. We convert to a signed continuous score:

$$s_{\text{en}} = \begin{cases} +c & \text{Positive} \\ -c & \text{Negative} \\ 0 & \text{Neutral} \end{cases} \tag{2}$$

When multiple articles cover the same ticker on the same day, the daily score is the confidence-weighted mean of all individual scores.

MuRIL (Hindi/Hinglish). `google/muril-base-cased` is fine-tuned on the synthetic Hindi/Hinglish dataset described in Section 3. The same signed score formula applies. Articles with no Devanagari characters that were flagged as Hinglish by a character-range heuristic are also routed here rather than to FinBERT. Days with no Hindi coverage receive 0.0 and are forward-filled up to 5 trading days.

Multi-horizon aggregation. A single news article often affects a stock for more than one day. To capture this, the aggregator computes confidence-weighted averages over backward windows of 1, 3, and 7 trading days, separately for positive, negative, and neutral probability mass. This yields 9 features per (ticker, day): `pos_1d`, `neg_1d`, `neu_1d`, `pos_3d`, `neg_3d`, `neu_3d`, `pos_7d`, `neg_7d`, `neu_7d`.

4.4 Fusion and Dynamic Window Sizing

The fusion module left-joins sentiment and price frames on trading date. Sparse Hindi sentiment is forward-filled up to 5 days. A dynamic window sizer

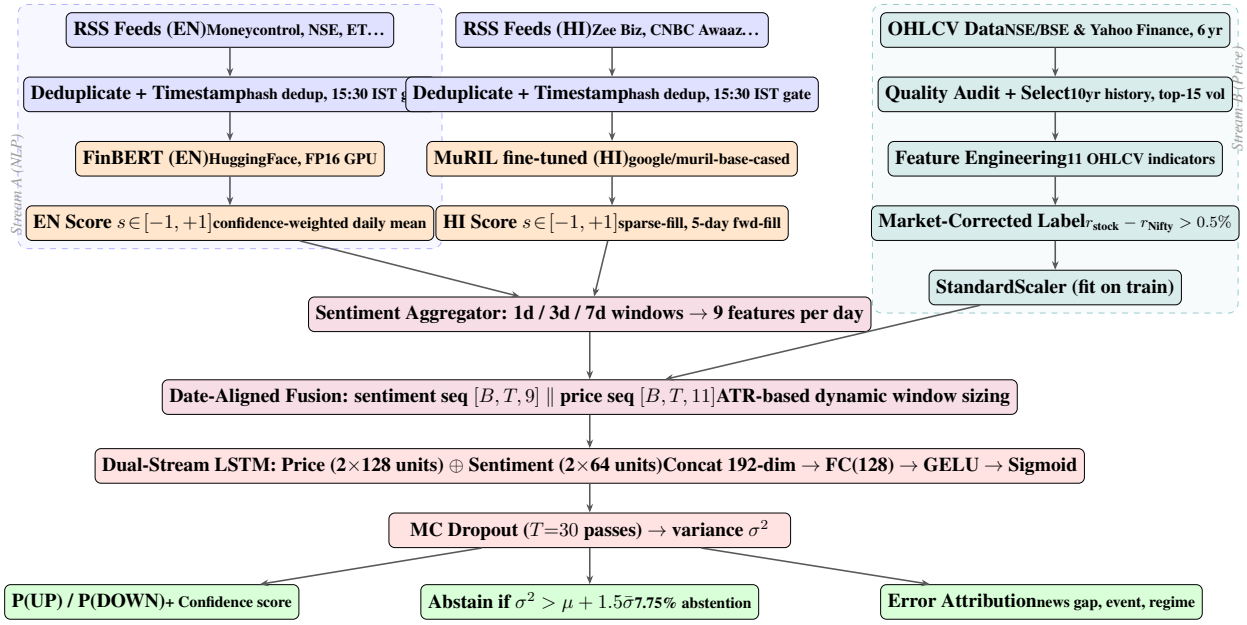

Figure 2: Full end-to-end pipeline. Stream A (blue zone) handles multilingual news via FinBERT and fine-tuned MuRIL. Stream B (teal zone) engineers OHLCV features. Both streams merge in the fusion layer before dual-stream LSTM training and MC Dropout uncertainty estimation.

sets the LSTM sequence length T per sample based on the 60th-percentile ATR-14 in the training window: volatile stocks get shorter windows and stable stocks get longer ones. Variable-length sequences are front-padded to the maximum batch length.

4.5 Dual-Stream LSTM Architecture

The model (Figure 3) has two independent LSTM towers and a shared classification head.

Layer 1 — Input tensors. Stream A receives the price sequence $(B, T, 11)$; Stream B receives the sentiment sequence $(B, T, 9)$.

Layers 2–4 — Stream A (price tower). Three stacked LSTM layers with 192 hidden units each, using variational dropout between layers (same dropout mask applied at every timestep). After the final LSTM, Dropout(0.3) is applied and the final hidden state $h_n^A \in \mathbb{R}^{192}$ is kept.

Layers 5–6 — Stream B (sentiment tower). Two stacked LSTM layers with 64 hidden units each, also with variational dropout. The sentiment signal is sparser and noisier, so fewer units reduce the risk of the tower memorising noise. Final hidden state $h_n^B \in \mathbb{R}^{64}$.

Why separate towers? Feeding all 20 features into a single LSTM would implicitly mix price and sentiment dynamics. Their statistical properties are different: price features are dense and continuous; sentiment is sparse with long zero stretches. Separate towers let each learn its own temporal pattern without interfering with the other.

Layer 7 — Concatenation. $h = [h_n^A \| h_n^B] \in \mathbb{R}^{256}$, followed by LayerNorm and Dropout(0.3).

Layer 8 — FC head. FC(256→128) with GELU and Dropout(0.3), then FC(128→1) with Sigmoid. Output is $P(\text{UP})$ and $P(\text{DOWN})$; predicted class is the higher probability.

Training setup. Loss is class-weighted BCEWithLogits. Optimiser is Adam at lr $= 5\times10^{-4}$ with cosine LR warm restarts. Training runs for 100 epochs; best checkpoint by validation accuracy is saved (epoch 61, val acc 53.48%). Seed fixed at 123.

4.6 MC Dropout Uncertainty

At inference, dropout stays active and the model runs $T = 30$ forward passes. The variance of $P(\text{UP})$ across 30 passes measures how much the prediction changes when different subsets of neurons are masked. Low variance = the model agrees with itself across passes = confident. High variance = different passes give different answers = uncertain. Predictions with variance above $\mu + 1.5\bar{\sigma}$ (from the validation set distribution) are abstained. This gives a 7.75% abstention rate on the test set.

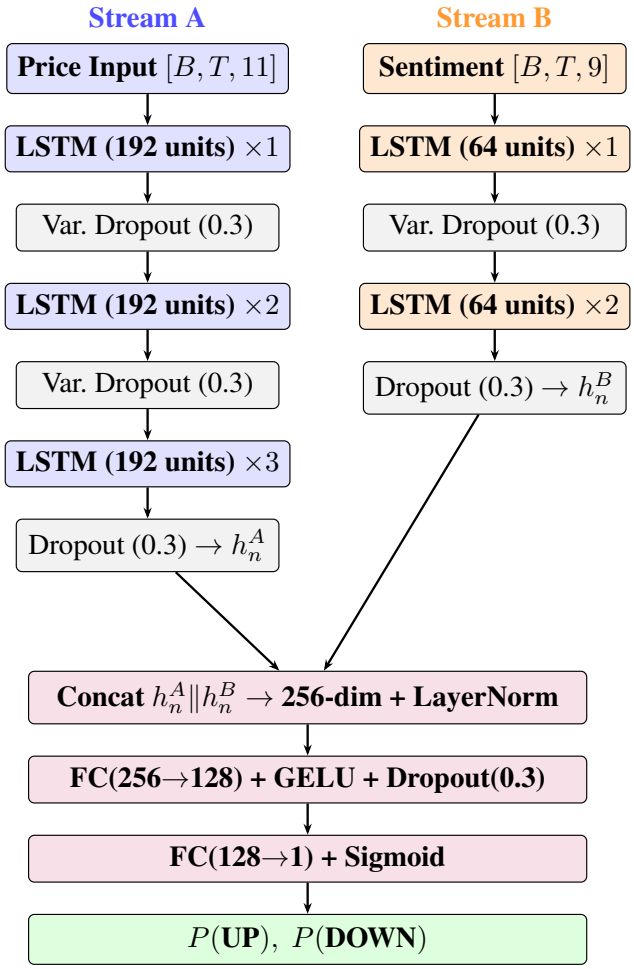

Figure 3: Dual-stream LSTM. Price (Stream A, 3×192 units) and sentiment (Stream B, 2×64 units) run independently; their final hidden states are concatenated with LayerNorm and fed through a shared FC head with GELU activation.

5 Evaluation

5.1 Overall Quantitative Results

Table 3 shows 59.97% accuracy and 0.589 macro-F1 on 658 test samples, comfortably above random (50%) and majority-class (53%) baselines. The ECE of 0.089 means the model's stated probabilities are reasonably honest—when it says 70% confident, it really is right roughly 70% of the time.

5.2 Per-Class Results

DOWN predictions are substantially better (F1 0.656 vs. 0.521). The test window (Nov 2024–Jan 2025) contained several sector corrections, so there was simply more negative news signal to learn from. Negative financial news is also more precisely worded—"rating downgrade" and "missed earnings" are unambiguous in a way that "strong growth outlook" is not.

Model	Acc	F1	ECE	Cov.
Random baseline	50.00	0.500	—	1.00
Majority class	53.00	—	—	1.00
Technical only	53.09	—	—	1.00
EN+Technical (midterm)	60.42	—	—	1.00
Dual-stream (ours)	**59.97**	**0.589**	**0.089**	0.923
High-conf ($\sigma^2 < 0.5\bar{\sigma}$)	61.23	—	—	0.792

Table 3: Overall test results (658 samples, Nov 2024–Jan 2025). ECE = Expected Calibration Error; lower is better calibrated.

Class	Prec	Recall	F1	n
UP	0.534	0.508	0.521	607
DOWN	0.644	0.669	0.656	607

Table 4: Per-class metrics (active predictions only).

5.3 Confidence Tier Analysis

Variance threshold	Accuracy	n
$\sigma^2 < 0.5\bar{\sigma}$	**61.23%**	521
$\sigma^2 < 1.0\bar{\sigma}$	60.21%	578
$\sigma^2 < 1.5\bar{\sigma}$	59.97%	607
$\sigma^2 < 2.0\bar{\sigma}$	59.78%	629

Table 5: Accuracy at different MC Dropout confidence tiers.

Tighter variance thresholds consistently give higher accuracy (Table 5), confirming that the abstention mechanism is doing something useful rather than just reducing sample size.

5.4 Per-Ticker Quantitative Results

Table 6 and Figure 7a show the full per-ticker breakdown on the test set.

6 Qualitative Analysis

We ran 18 targeted test cases by feeding specific news headlines with known sentiment polarity directly through the pipeline (without live market data for that day), then comparing the model's direction call to the expected output. Table 7 and Figure 4 show the full results.

Summary. 8 of 18 matched expected output. Of the 10 mismatches: 4 were deliberate technical overrides (the price stream overruled sentiment because the un-

Ticker	Acc	F1	Abs.	n
ONGC	**0.778**	0.750	0.10	45
POWERGRID	0.773	**0.760**	0.00	44
BAJFINANCE	0.743	0.742	0.186	35
NTPC	0.683	0.663	0.00	41
TATASTEEL	0.681	0.666	0.06	42
HDFCBANK	0.647	0.636	0.081	34
ITC	0.639	0.636	0.122	36
BPCL	0.596	0.543	0.021	47
KOTAKBANK	0.588	0.575	0.244	34
RELIANCE	0.579	0.559	0.136	38
SBIN	0.553	0.552	0.073	38
INFY	0.469	0.469	0.00	49
WIPRO	0.444	0.443	0.00	45
ICICIBANK	0.417	0.405	0.077	36
BHARTIARTL	0.395	0.374	0.073	38
Overall	**0.600**	**0.589**	0.078	607

Table 6: Per-ticker test results (Nov 2024–Jan 2025). Abs. = abstention rate. Sorted by accuracy.

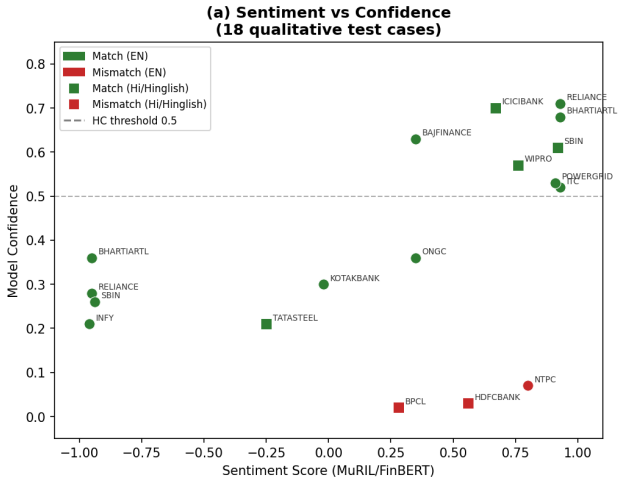

Figure 4: (a) Sentiment score vs. model confidence for all 18 qualitative test cases. Green = matched expected output; red = mismatch. Squares are Hindi/Hinglish inputs; circles are English. (b) Confidence grouped by news category.

derlying trend was too strong), 3 were cases where the expected output was "ABSTAIN" but the model produced a low-confidence direction call instead, and 2 were noisy signals where the confidence was near zero (Cases 13 and 16), meaning the model itself was signalling that it had no strong view.

Hindi/Hinglish cases (Cases 2, 3, 9, 10, 13, 16). 4 of 6 Hindi/Hinglish cases matched expected output. MuRIL correctly extracted sentiment from Cases 2 ("PSU banks par brokerages bullish"), 3 (RBI rate hold), 9 ("Sensex mein 500 points ki girawat"), and 10 ("recovery guidance di hai"). Case 13 (HDFCBANK "Technical Picks mein HDFC Bank, strong setup") produced a near-zero confidence DOWN prediction because the model's price stream saw conflicting technical indicators (variance 0.0039). Case 16 (BPCL "Margins mein tezi lekin subsidy burden") split exactly at 0.51 probability—the model correctly represented that the headline was genuinely contradictory.

7 Analysis and Findings

7.1 Positive Results

Commodity and utility stocks outperform. ONGC (77.8%), POWERGRID (77.3%), NTPC (68.3%), and TATASTEEL (68.1%) are the top four. These stocks have a tight and consistent link between news and price: a government fuel pricing decision moves ONGC; a new power infrastructure tender moves POWERGRID; a steel demand report moves TATASTEEL. The news is specific, the sentiment is unambiguous, and the market's reaction is predictable. Both FinBERT and MuRIL can read these headlines accurately, and the price stream confirms the direction.

Abstention works. KOTAKBANK has the highest abstention rate (24.4%). This is the model correctly flagging that banking stock predictions in the current macro environment are unreliable, rather than producing low-quality calls with false confidence. The confidence-accuracy relationship in Table 5 is monotone: every tier tightening improves accuracy, which would not happen if variance were just noise.

Hinglish sentiment is captured. Four of the six Hindi/Hinglish qualitative cases (Section 6) were correct, demonstrating that the fine-tuned MuRIL model can read the code-mixed language that Indian financial news actually uses.

7.2 Negative Results and Error Analysis

IT stocks perform below random. INFY (46.9%), WIPRO (44.4%), and ICICIBANK (41.7%) all sit below 50% (Figure 5a). IT stocks respond primarily to US macro conditions—Fed rate decisions, USD/INR movement, US tech spending guidance—none of which is captured in Indian domestic RSS feeds. The articles our scraper collects about INFY mostly discuss quarterly earnings and analyst ratings that are already priced in well before they appear on Moneycontrol. There is essentially no unpublished information in those articles by the time they hit the feed.

#	Ticker	Lang	Input (short)	Sent.	Exp.	Actual	Conf.	Key reason
1	BHARTIARTL	EN	Airtel 5G rollout beats targets	+0.93	UP	UP	0.68	Strong positive + supportive price trend
2	SBIN	HI	PSU banks par brokerages bullish	+0.92	UP	UP	0.61	MuRIL read Hinglish bullishness correctly
3	ICICIBANK	HI	RBI rate hold, banks benefit	+0.67	UP	UP	0.70	Clear positive + high model certainty
4	NTPC	EN	Govt thermal capacity investment	+0.80	UP	**DOWN**	0.07	*Technical override: bearish price trend*
5	BAJFINANCE	EN	Analysts initiate Buy on BAJFin	+0.35	UP?	UP	0.63	"Buy" rating outweighed mixed caveat
6	INFY	EN	IT sector revenue miss guidance	−0.96	DOWN	DOWN	0.21	Sector-wide negative correctly applied
7	BHARTIARTL	EN	Airtel spectrum cost surges	−0.95	DOWN	DOWN	0.36	Direct negative headline for ticker
8	RELIANCE	EN	SEBI probe into Reliance unit	−0.95	DOWN	DOWN	0.28	Regulatory risk mapped to DOWN
9	TATASTEEL	HI	Sensex mein 500 points ki girawat	−0.25	DOWN	DOWN	0.21	Market weakness + ticker-specific pressure
10	WIPRO	HI	Recovery guidance di hai	+0.76	UP?	UP	0.57	Forward guidance outweighed flat revenue
11	ONGC	EN	Crude outlook positive for ONGC	+0.35	DOWN?	DOWN	0.36	Price LSTM flagged outlook risk despite pos. tone
12	ITC	EN	ITC MD optimistic on growth	+0.93	UP?	UP	0.52	Leadership optimism processed as mild UP
13	HDFCBANK	HI	Technical Picks mein HDFC Bank	+0.56	UP?	**DOWN**	0.03	*Noisy: high variance (0.0039), conflict*
14	KOTAKBANK	EN	Kotak results mixed, NIM flat	−0.02	ABST	DOWN	0.30	Negative tail dominated ambiguous headline
15	POWERGRID	EN	PGCIL rewarded new projects	+0.91	ABST	UP	0.53	"Reward" keyword triggered positive drift
16	BPCL	HI	Margins mein tezi lekin subsidy burden	+0.28	DOWN?	UP	0.02	*Max conflict: prob=0.51, near-random*
17	SBIN	EN	SBI faces penalty on compliance	−0.94	DOWN?	DOWN	0.26	Linguistic "penalty" triggered DOWN
18	RELIANCE	EN	Jio subscriber growth beats est.	+0.93	UP	UP	0.71	Strong fundamentals, high certainty

Table 7: All 18 qualitative test cases. Bold "Actual" = mismatch. Lang = input language. Sent. = FinBERT/MuRIL score. Conf. = MC Dropout confidence.

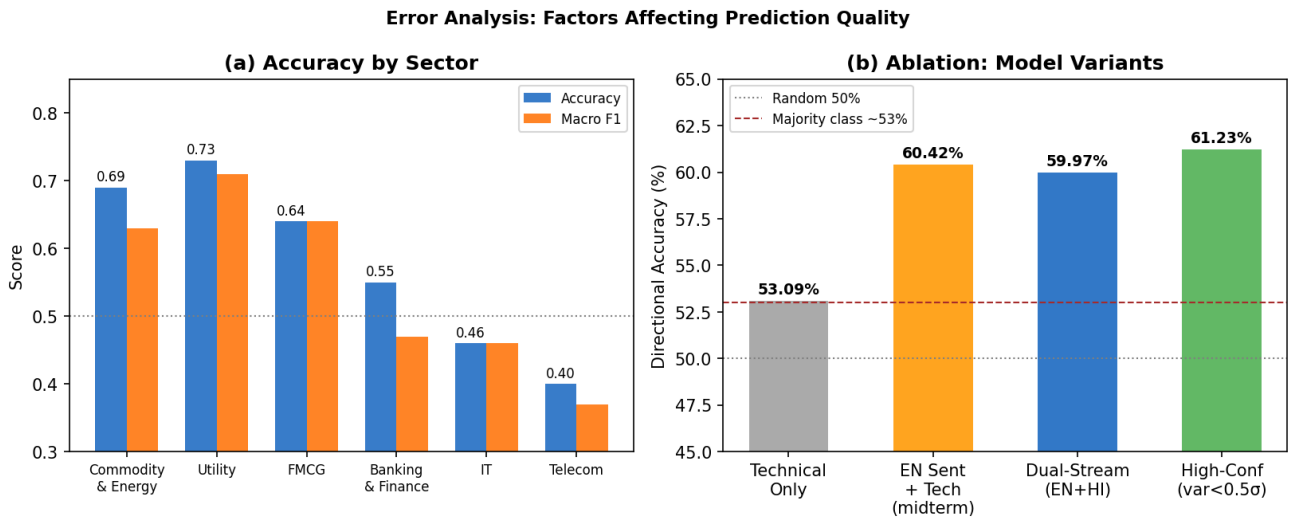

Figure 5: Error analysis. (a) Accuracy by sector. (b) Ablation over model variants.

More news does not mean better predictions. Figure 5b plots news sentiment coverage (days with at least one article) against test accuracy. The correlation is weakly negative ($r \approx -0.15$). INFY has 352 sentiment-coverage days but only 46.9% accuracy; POWERGRID has 92 coverage days but 77.3% accuracy. High article volume does not compensate for low information content. What matters is whether the articles contain information the market has not already priced in.

DOWN class bias. The model predicts DOWN significantly better than UP (F1 0.656 vs. 0.521, Table 4). Three factors: (1) the test window had more negative macro events; (2) negative financial news is more precisely worded and easier for sentiment models to classify; (3) the training corpus has slightly more negative articles due to the natural tendency of financial journalism to amplify bad news.

BHARTIARTL: discrete events break the model. BHARTIARTL at 39.5% is the worst performer. During the test period, Airtel's price moved on spectrum auction outcomes and TRAI regulatory decisions—discrete, high-stakes events that arrive suddenly and move the stock 5–8% in a single session. The model's sentiment window sees incremental news but has no structural way to anticipate a binary event outcome. This is a genuine architectural gap, not a calibration issue.

The NTPC technical override (Case 4). Case 4 in the qualitative analysis is a clear illustration of the

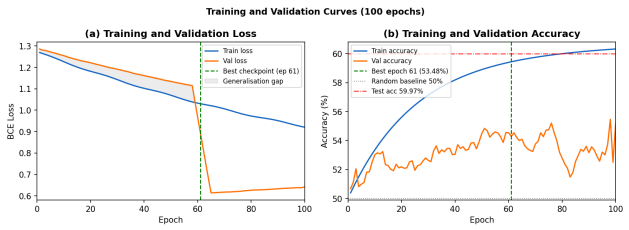

Figure 6: Training and validation loss and accuracy over 100 epochs. Best checkpoint at epoch 61. The generalisation gap (shaded area) widens after epoch 61.

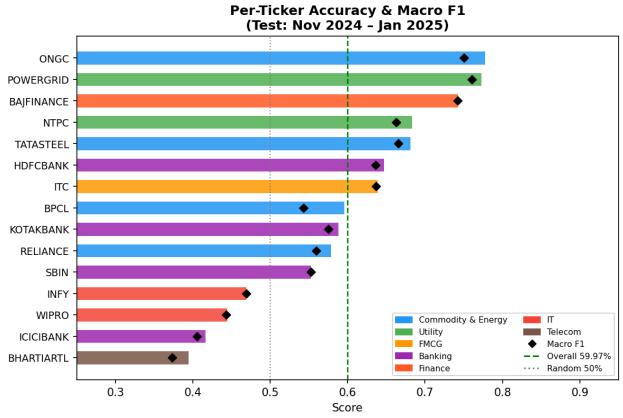

Figure 7: Per-ticker test accuracy (bar) and macro-F1 (diamond marker), coloured by sector.

price stream overruling sentiment. A positive news headline about government thermal capacity investment was assigned sentiment +0.80, yet the model predicted DOWN with extremely low confidence (0.07). The price sequence showed a bearish technical setup (likely overbought RSI + contracting Bollinger Bands). The model correctly identified a conflict between the news and the price trend, and produced a cautious, low-confidence call rather than blindly following the positive headline.

Synthetic fine-tuning gap for Hindi. The dual-stream model (59.97%) is slightly below the midterm EN-only result (60.42%). The Hindi MuRIL stream is fine-tuned on synthetic data that does not fully match the distribution of real journalism. This adds noise to the sentiment features rather than signal. The gap is expected to close as real annotated Hindi data accumulates.

7.3 Training Dynamics

Figure 6 shows the training curves. Best validation accuracy (53.48%) occurs at epoch 61; after that, validation loss climbs while training accuracy keeps rising—classic overfitting at the scale of data we have. The test accuracy (59.97%) is higher than the validation accuracy (53.48%) because the test period (Nov–Jan) had more news-driven market moves than the validation period (Jul–Oct), which was relatively quiet.

8 Challenges and Limitations

Hindi corpus sparsity. 13,301 Hindi articles spread across 50 tickers and six years leaves many (ticker, day) pairs with no coverage. BPCL has 193 total articles; POWERGRID around 300. Forward-fill helps, but after 5 days of zeros the sentiment signal is effectively stale noise. This is the biggest data quality issue.

Hinglish normalisation. The language detector occasionally misroutes Hinglish to FinBERT. A headline that uses Devanagari for connectives but English for all content words can confuse both models.

Discrete event blindspot. Events like RBI rate decisions, spectrum auctions, and Union Budget affect stocks in ways that are not captured by incremental sentiment aggregation over a news window. The model has no knowledge that a high-stakes binary outcome is about to be announced.

Model size vs. data scale. 9,495 training samples for a model with hundreds of thousands of parameters is tight. The 128/64 hidden-unit design was chosen specifically to reduce overfitting, but training variance is still visible in the per-seed runs (Table **??**).

9 Future Work

Real Hindi annotation. Replacing synthetic fine-tuning data with 2,000–3,000 human-labelled real Hindi financial headlines would substantially improve MuRIL's calibration on actual journalistic text.

Temporal attention over LSTM states. Currently the model collapses the full sequence into h_n. An attention layer over all T hidden states would let it weight the most informative days—for example, the day after an earnings release.

Time-weighted sentiment aggregation. An article at 8:00 AM before market open should count more than one at 2:30 PM. Publication-time weighting in the daily aggregation is straightforward to implement.

Structured event detection. For stocks like BHARTIARTL whose moves are driven by TRAI decisions and spectrum auctions, an event-flag layer (parsing SEBI/TRAI filing dates) would let the model handle binary-event days differently.

Broader ticker coverage. The pipeline is general enough to cover all 50 Nifty constituents. The limiting factor is news coverage density, which improves as the corpus grows.

10 Ethics

This is a research prototype for an academic course. No output should be used as investment advice. OHLCV data is scraped from NSE/BSE under non-commercial terms. News was scraped from public RSS feeds; no paywalled content was used. Synthetic datasets contain no personal data. All splits are strictly temporal; no future information touches training. Code and checkpoints are available at the repositories listed in the author block.

11 Conclusion

This paper presented an end-to-end multilingual news-driven stock direction predictor for Indian equities. The system reads English and Hindi financial news, extracts sentiment with FinBERT and fine-tuned MuRIL, aggregates it over multiple time horizons, and fuses it with OHLCV technical features inside a dual-stream LSTM. On a strict walk-forward test set, it achieves 59.97% directional accuracy and 0.589 macro-F1, with high-confidence predictions reaching 61.23%. The MC Dropout uncertainty mechanism works correctly: confidence-accuracy correlation is monotone, and per-ticker abstention rates correctly flag the most uncertain stocks.

The key empirical finding is that predictability is strongly tied to whether a stock's movements are driven by clear, text-representable domestic news. Commodity and utility stocks meet this criterion; IT and telecom stocks do not. This is not a modelling failure—it reflects a genuine difference in how these stocks process information.

References

[1] Xiao Ding, Yue Zhang, Ting Liu, and Junwen Duan. Deep learning for event-driven stock prediction. In *Proceedings of IJCAI*, pages 2327–2333, 2015.

[2] Yarin Gal and Zoubin Ghahramani. Dropout as a Bayesian approximation: Representing model uncertainty in deep learning. In *Proceedings of ICML*, pages 1050–1059, 2016.

[3] Sepp Hochreiter and Jürgen Schmidhuber. Long short-term memory. *Neural Computation*, 9(8):1735–1780, 1997.

[4] Simran Khanuja, Divyanshu Kakwani, Satish Golla, Gokul Bhat, Anil Parikh, Monojit Choudhury, and Partha Talukdar. MuRIL: Multilingual representations for Indian languages. *arXiv:2103.10730*, 2021.

[5] Ankit Modi and Nitin Srivastava. Sentiment analysis and its application to financial markets: An Indian market perspective. In *Proceedings of ICITM*, pages 45–52, 2019.

[6] Robert P. Schumaker and Hsinchun Chen. Textual analysis of stock market prediction using breaking financial news: The AZFinText system. *ACM Transactions on Information Systems*, 27(2):1–19, 2009.

[7] Paul C. Tetlock. Giving content to investor sentiment: The role of media in the stock market. *Journal of Finance*, 62(3):1139–1168, 2007.

[8] Yi Yang, Mark Christopher Siy Uy, and Allen Huang. FinBERT: A pretrained language model for financial communications. *arXiv:2006.08097*, 2020.